Background and Theoretical Foundations

Active Inference represents a paradigm shift in our understanding of cognition, perception, and action. Originating from the Free Energy Principle [**?**], Active Inference provides a unified mathematical formalism for understanding biological agents as systems that minimize variational free energy through perception and action (see Figure 1). Recent advances have extended Active Inference to scale-free formulations [**?**] and variational planning [**?**], while metacognitive architectures [**?**, **?**] have demonstrated the practical applicability of these principles to AI systems. This section establishes the theoretical foundations that enable Active Inference to operate as a meta-theoretical methodology—specifying the frameworks within which cognition occurs.

![Active Inference Concepts](../output/figures/perception$_action_loop.png)

Figure 1: Core concepts in Active Inference showing the relationship between perception, action, and free energy minimization. Active Inference unifies perception (inferring hidden states from observations) and action (selecting behaviors that minimize expected free energy)